

The shortest path between two points isn't always a line

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December 20, 2023

When we first studied geometry, we learned that the shortest path between two points is a line segment, usually of the form $y(x) = mx + b$. This is a perfectly valid statement but only under the constraint that the space *must* be Euclidean. In other words, this condition only holds for spaces that are globally Cartesian (more formally, like $\mathbb{R}^n$), and the usual Pythagorean formula holds for any distance measured, that is,

$$s^2 = x^2 + y^2, \quad ds^2 = dx^2 + dy^2 \quad (1)$$

for arbitrarily small displacements in the x and y directions. Moving into non-Euclidean regimes, the way we measure distance changes drastically and can depend on how one constructs a coordinate system. How, then, might we develop a way to find the shortest distance between two points in an arbitrary space?

From here on, we will assume a knowledge of calculus and spherical coordinates in an example later on.

Developing the Euler-Lagrange equations

There are a few approaches to finding expressions for the shortest distance between two points in an arbitrary space, but I've opted to use a method from the calculus of variations. As the name suggests, this method deals with variations in functions to arrive at an optimal solution. The basic premise is that we use a mapping known as a **functional** and extremize it to arrive at a desired solution. This functional, unlike your usual function, takes functions as its input and spits out a scalar – we can therefore write the mapping like $S : \mathcal{F} \rightarrow \mathbb{R}$, where $\mathcal{F}$ is the set of our functions we wish to use as inputs (function space) and S is our functional mapping. I've opted to use S to foreshadow the purpose of using functionals in this case. The actual expression is, in general,

$$S[\sigma] = \int_{\xi_1}^{\xi_2} L(\xi, \sigma, \sigma') d\xi, \quad \text{where } \sigma' \equiv \frac{d\sigma}{d\xi}. \quad (2)$$

$L(\xi, \sigma, \sigma')$ is our function input, and extremizing $S[\sigma]$ will involve $L(\xi, \sigma, \sigma')$ in some way, to be clarified later. The goal from here is to find conditions that are satisfied upon minimizing $S[\sigma]$. The variables in L are related such that $\sigma = \sigma(\xi)$.

The motivation here is that we will treat $S[\sigma]$ as distance in an arbitrary space and find out what happens when we minimize it, thus finding an expression for the shortest distance between two points in any space. But how exactly do we minimize $S[\sigma]$?

A perturbative approach in σ .

Here is where some magic happens. Remember that σ is a function of ξ , but *a priori* it is unknown. This gives us the freedom to define it such that

$$\sigma(\xi) = f(\xi) + \varepsilon \eta(\xi), \quad (3)$$

where ε is our perturbative parameter (we will assume it is small). $f(\xi)$ is taken to be a function that will minimize $S[\sigma]$ and $\eta(\xi)$ will be our perturbation function. Since it is a perturbation, we expect that it must disappear at the extremal points, where $\eta(\xi_1) = \eta(\xi_2) = 0$. We also see that as we change ε , $S[\sigma]$ changes. At the optimal function $\sigma(\xi) = f(\xi)$ that extremizes $S[\sigma]$, $\varepsilon = 0$, and these conditions imply that

$$\left. \frac{dS}{d\varepsilon} \right|_{\varepsilon=0} = 0 \quad (4)$$

To get further information from this, we utilize the multivariable chain rule to write the exact differential dS to help us with equation 4. We note that $S = S(\sigma, \sigma')$ because we have integrated over ξ and as a result remove that variable from the resulting expression. Hence,

$$dS = \frac{\partial S}{\partial \sigma} d\sigma + \frac{\partial S}{\partial \sigma'} d\sigma' \quad (5)$$

and as a result

$$\frac{dS}{d\varepsilon} = \frac{\partial S}{\partial \sigma} \frac{d\sigma}{d\varepsilon} + \frac{\partial S}{\partial \sigma'} \frac{d\sigma'}{d\varepsilon}. \quad (6)$$

Here we can recognize that because

$$\sigma = f(\xi) + \varepsilon \eta(\xi), \quad \frac{d\sigma}{d\xi} = \frac{df}{d\xi} + \varepsilon \frac{d\eta}{d\xi},$$

we can write

$$\frac{d\sigma}{d\varepsilon} = \eta(\xi), \quad \frac{d\sigma'}{d\varepsilon} = \eta'(\xi) \text{ where } \eta'(\xi) \equiv \frac{d\eta}{d\xi}. \quad (7)$$

We can also evaluate the partial derivatives:

$$\frac{\partial S}{\partial \sigma} = \int_{\xi_1}^{\xi_2} \frac{\partial L}{\partial \sigma} d\xi, \quad \frac{\partial S}{\partial \sigma'} = \int_{\xi_1}^{\xi_2} \frac{\partial L}{\partial \sigma'} d\xi \quad (8)$$

Substituting equations 7 and 8 into equation 6, we have

$$\frac{dS}{d\varepsilon} = \int_{\xi_1}^{\xi_2} \left(\frac{\partial L}{\partial \sigma} \eta + \frac{\partial L}{\partial \sigma'} \eta' \right) d\xi \quad (9)$$

This is some good progress! However, to further evaluate this, we need to know more. We now find a way to rewrite the integrand into something more revealing. Let's integrate the second term in the integrand by parts:

$$\begin{aligned} \text{Since } \int f dg = f g - \int g df, \text{ let } f = \frac{\partial L}{\partial \sigma'}, dg = \eta' d\xi = d\eta. \text{ Then,} \\ \int_{\xi_1}^{\xi_2} \frac{\partial L}{\partial \sigma'} \eta' d\xi = \underbrace{\left. \frac{\partial L}{\partial \sigma'} \eta \right|_{\xi_1}^{\xi_2}}_{\text{since } \eta(\xi_1)=\eta(\xi_2)=0} - \int_{\xi_1}^{\xi_2} \eta \frac{d}{d\xi} \left(\frac{\partial L}{\partial \sigma'} \right) d\xi. \end{aligned}$$

Let us substitute this back into equation 9 to get

$$\frac{dS}{d\varepsilon} = \int_{\xi_1}^{\xi_2} \eta(\xi) \left(\frac{\partial L}{\partial \sigma} - \frac{d}{d\xi} \frac{\partial L}{\partial \sigma'} \right) d\xi \quad (10)$$

Now we make a crucial step. Since we noted earlier that $\left. \frac{dS}{d\varepsilon} \right|_{\varepsilon=0}=0$, it must also follow that

$$0 = \left. \frac{dS}{d\varepsilon} \right|_{\varepsilon=0} = \int_{\xi_1}^{\xi_2} \eta \left(\frac{\partial L}{\partial \sigma} - \frac{d}{d\xi} \frac{\partial L}{\partial \sigma'} \right) d\xi.$$

Because we never made any assumptions about $\eta(\xi)$ other than its boundary conditions, it is completely arbitrary. But the integral always yields zero. What that means is that the difference in parentheses must be identically zero for all ξ between ξ_1 and ξ_2 ! This allows us to write the well-known **Euler-Lagrange equations**

$$\frac{\partial L}{\partial \sigma} - \frac{d}{d\xi} \frac{\partial L}{\partial \sigma'} = 0 \quad (11)$$

which allows us to place a powerful constraint on the extremizing function $L(\xi, \sigma, \sigma')$. Using the Euler-Lagrange equations will usually give us a useful differential equation that, in this case of distance, allows us to write an equation of one parametrizing spatial variable in terms of others. Let's look at two examples, one Euclidean and one non-Euclidean.

The shortest path in Euclidean space

We use the above approach to explicitly show that the shortest path between two points in Euclidean space is a line segment. Remember that, for our purposes, we treat $S[\sigma]$ as path length and we aim to minimize it. If we let our path be denoted γ , we can just add up (read: integrate) differential path length elements to get the total path length:

$$S = \int_{\gamma} ds. \quad (12)$$

We can get a more explicit form of dS by using equation 1:

$$ds^2 = dx^2 + dy^2 \quad \implies \quad ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + (y')^2} dx, \text{ where } y' \equiv \frac{dy}{dx}.$$

If we read off the integrand and match it to our general form above, we get that

$$L(x, y, y') = \sqrt{1 + (y')^2} \text{ since } S = \int_{x_1}^{x_2} \sqrt{1 + (y')^2} dx.$$

Rewriting the Euler-Lagrange equations to fit our situation, we first see that

$$\frac{\partial L}{\partial y} = 0, \quad \frac{\partial L}{\partial y'} = \frac{y'}{\sqrt{1 + (y')^2}}.$$

Since its ε derivative is zero, it implies that $\partial L / \partial y' = c$, where c is some constant. We can then isolate variables to get

$$y' = \frac{c}{1 - c^2} \quad (13)$$

Equation 13 reduces to a separable differential equation, where we can now define $m \equiv \frac{c}{1 - c^2}$. Then

$$\begin{aligned} \frac{dy}{dx} &= m \\ \int dy &= m \int dx \\ y - y_0 &= m x, \end{aligned}$$

where y_0 is a constant of integration. If we redefine y_0 as b , we immediately see that our function is, in fact, a line segment:

$$y = m x + b \quad (14)$$

What about on a sphere?

We now move onto a non-Euclidean case. While the surface of a sphere locally resembles two-dimensional Euclidean space, $\mathbb{R}^2$, globally, it is clearly not Euclidean. Like the Euclidean example, we need a way to define the differential path length element. This is not as straightforward, but we can derive it nonetheless. Due to its local similarity to Euclidean 2D space, we can still apply the Pythagorean theorem for arbitrarily small displacements. If we imagine a random point on this sphere, which we denote as (θ, ϕ) , we can procure this mental image:

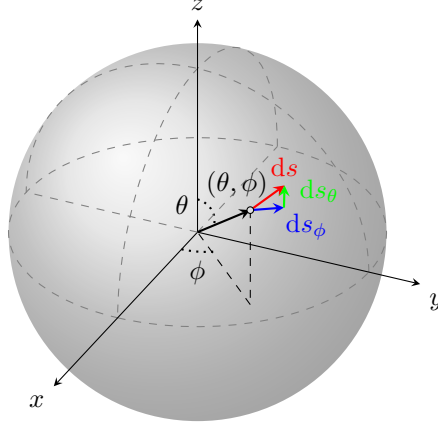


Figure 1: A way to visualize an arbitrarily small displacement on a sphere.

If we zoom in to the triangle with ds , ds_θ , and ds_ϕ , we can use the Pythagorean theorem and conclude

$$ds^2 = ds_\phi^2 + ds_\theta^2.$$

But we'd like to express this equation in terms of differential displacements in θ and ϕ so we can actually use the Euler-Lagrange equations from above. Inspecting the sphere and its displacements, we see that for a sphere of radius r , ds_θ is simply a small displacement in the great circle of the sphere, so that $ds_\theta = r d\theta$. This is not the case for ds_ϕ , since we should note that the direction ds_ϕ goes *does not lie on the path of a great circle*. Instead, it lies on a circle whose radius depends on θ . To see this more clearly, consider the extreme case where the polar angle $\theta = 0$. In such a case, no matter what we do, we cannot obtain a nonzero ds by only changing ϕ , since that latitudinal set lies on exactly one point, which by definition has no length. Instead, the circle whose path ds_ϕ follows has a radius of $r \sin \theta$, which can be seen by taking the component of our (θ, ϕ) coordinate parallel to the xy plane, yielding $r \sin \theta$. Thus,

$$ds^2 = r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2. \quad (15)$$

With this in mind, we proceed much like the Euclidean case. In this case, our path γ connects two points on the sphere, which we will call (θ_1, ϕ_1) and (θ_2, ϕ_2) . We see that this time, our differential path length element is

$$ds = r \sqrt{d\theta^2 + \sin^2 \theta d\phi^2} = r \sqrt{1 + \sin^2 \theta (\phi')^2} d\theta, \text{ where } \phi' \equiv \frac{d\phi}{d\theta}. \quad (16)$$

By the same logic as above (you can verify this for yourself), we then have that

$$L(\theta, \phi, \phi') = r \sqrt{1 + \sin^2 \theta (\phi')^2},$$

and we integrate over θ from θ_1 to θ_2 . We see that we can once again use the Euler-Lagrange equations, observing that

$$\frac{\partial L}{\partial \phi} = 0 \text{ and } \frac{\partial L}{\partial \phi'} = \frac{r \phi' \sin^2 \theta}{(1 + \sin^2 \theta (\phi')^2)^{1/2}}.$$

Since $\partial L/\partial \phi$ is zero, we see that

$$-\frac{d}{d\theta} \left(\frac{\partial L}{\partial \phi'} \right) = 0,$$

implying that $\partial L/\partial \phi'$ is a constant, which I will choose to denote as c . This allows us to derive an expression for ϕ' , which will then give us a separable differential equation and allow us to solve for $\phi(\theta)$. In the following steps, I have gotten rid of r because we have assumed a sphere of constant radius.

$$\begin{aligned} \frac{\partial L}{\partial \phi'} = c &= \frac{\phi' \sin^2 \theta}{\sqrt{1 + \sin^2 \theta (\phi')^2}} \\ \implies \phi &= \int \frac{c}{\sin \theta \sqrt{\sin^2 \theta - c^2}} d\theta. \end{aligned}$$

Unfortunately, this integral is a bit of a pain to evaluate, but there's a nice u -substitution we can make utilizing the trig identity $\csc^2 \theta = 1 + \cot^2 \theta$.

$$\begin{aligned} \frac{c}{\sin \theta \sqrt{\sin^2 \theta - c^2}} &= \frac{c}{\sin^2 \theta \sqrt{1 - c^2/\sin^2 \theta}} \\ &= \frac{c \csc^2 \theta}{\sqrt{1 - c^2 \csc^2 \theta}} \\ &= \frac{c \csc^2 \theta}{\sqrt{1 - c^2 (1 + \cot^2 \theta)}} \end{aligned}$$

Let $u = \cot \theta$, $du = -\csc^2 \theta d\theta$, so that

$$\begin{aligned} \int d\phi &= \int \frac{c \csc^2 \theta}{\sqrt{1 - c^2 - c^2 \cot^2 \theta}} = - \int \frac{c}{\sqrt{1 - c^2 - c^2 u^2}} du \\ &= - \int \frac{du}{\sqrt{a^2 - u^2}}, \text{ where } a^2 \equiv \frac{1 - c^2}{c^2} \end{aligned}$$

The right hand integral is seen to evaluate to the inverse cosine of u/a , and the left hand integrand gives $\phi + \phi_0$, where ϕ_0 is some constant of integration. This means that

$$\phi + \phi_0 = \cos^{-1} \left(\frac{u}{a} \right). \quad (17)$$

Upon replacing our u -substitution, equation 17 becomes

$$\cos(\phi + \phi_0) - \frac{1}{a} \cot \theta = 0.$$

Using the cosine sum of angles identity and multiplying through by $\sin \theta$, we have

$$0 = a \cos \phi_0 \sin \theta \cos \phi - a \sin \phi_0 \sin \theta \sin \phi - \cos \theta. \quad (18)$$

While this might initially seem meaningless, we can use our definitions of spherical coordinates and the fact that a and ϕ_0 are both arbitrary constants to conclude that this equation is actually secretly that of a plane in $\mathbb{R}^3$ that goes through the origin, namely that equation 18 is equivalent to

$$0 = Ax + By + Cz. \quad (19)$$

Because all intersections of the sphere and a plane that goes through the origin yield great circles, this proves that, indeed, the shortest path between any two points on a sphere must be that along a great circle.